ConnectBox

What is a ConnectBox?

ConnectBox is a self-contained, self-powered, Wi-Fi-based portable media distribution system. Programmed with an open-source software platform, it allows you to create a full 'digital library' with your own Gospel and educational media (text, video, audio, and applications). ConnectBox functions off-the-grid, without access to internet or electricity, and connects through the browser on any Wi-Fi enabled device. ConnectBox will support over 16+ simultaneous 360p video streams/download sessions from different connected devices.



Where would I use a ConnectBox?

ConnectBox allows you to provide unrestricted access to a virtually limitless variety of digital media. Users connect with devices they already own; you provide internet connection to a wealth of custom resources by uploading them to your unit.

What does ConnectBox cost?

Each ConnectBox can be built for the surprisingly low hardware cost starting at $70. With an internal uSD card of 64GB of storage space for the operating system and media you load onto the device. You can add an external USB flash drive up to 512GB formatted as FAT32, exFAT, NTFS or ext4. This external flash drive will override the media on the internal flash drive for quick media changes.



What's so unique about a ConnectBox?

- ConnectBox works without access to the internet.
- Streams and downloads movies, music, audio files, photos, text files and device applications—any content you upload.
- Opens in the phone's language when the ConnectBox has it, including right-to-left languages such as Arabic – the Wi-Fi welcome page as well as the menus.
  Menu wording for the 50 most spoken languages is built in, so no internet is needed.
- Languages can be defined by simply creating a folder with the language ISO code at the root file level. Media for that language then goes into that folder.
- No need to download special applications or purchase new equipment—use any Wi-Fi-capable device with its own built-in web browser.
- Integrated PDF, EPUB, Word (.docx and older .doc), Excel (.xlsx, .xls) and PowerPoint (.pptx and older .ppt) readers let phone users read these documents without downloading any special apps.
- Download buttons show the size of each file, so users know how big it is before they download it.

- Offline websites (ZIM files: Wikipedia, Stack Exchange, PhET, TED and more) open in a built-in reader with search.
- The user interface dynamically scales to the screen size of each device as it connects.
- Built-in Administration Interface allows you to customize the SSID, host name, Wi-Fi channel.
- Optionally shares the ConnectBox's own internet connection (cable or Wi-Fi) with the people connected to it – off unless the administrator turns it on.
- Plays a playlist of its media, with sound, on a monitor or TV (HDMI, Raspberry Pi models).
- The Administration Interface asks for its password every time, and the login ends as soon as you leave it, so a phone is never left logged in as administrator.
- You can even include your own organization's branding.
- Built in statistics allow you to track numbers of users and files downloaded/viewed. These can be synchronized to a host server that can track media usage over many different ConnectBoxes.
- Main menus include simple, icons or graphics derived from the media. Additionally, you can develop and insert more customized graphics as needed.
- Storage capacity is flexible, limited only by the size of flash drive attached via the built-in USB port.
- Media can be copied from a USB flash drive onto the ConnectBox with its front-panel buttons, with the progress shown on its display. A USB flash drive plugged in later is shown instead; when it is removed, the ConnectBox goes back to its own media. The display also shows its network addresses.
- ConnectBox creates its own Wi-Fi hotspot, eliminating any need for wires.
- Battery-powered: no need for AC mains, power adapters, or external batteries. Battery can be recharged using 5V micro-USB charger or optional external solar panels.
- All of this—and it still fits into your shirt pocket!

How do I get a ConnectBox?
ConnectBox can be purchased by contacting contact@globalsystems.design Additional information on how to customize or utilize the ConnectBox can be found on the website http://connectbox.org.

How do I set up a host system to manage and create content updates for field ConnectBoxes?
By contacting us we will work with you to set up the server for managing multiple ConnectBoxes in the field. You can use your own Ubuntu 24.04 server or a virtual server from AWS, Digital Ocean or another provider. Working with you or your IT team we can create the management system using a script to install the needed services. Then you can register the units in the field. Get the statistical information when your field unit manager attaches the ConnectBox to an internet connection via tethering or by Wi-Fi signal to the internet.

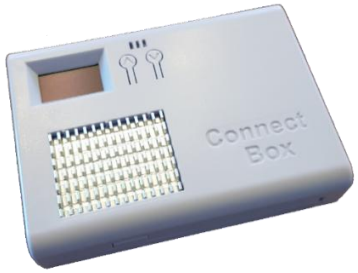
Other resources

Wi-Fi Media Boxes – Which One is for You?: https://tinyurl.com/wifimediaboxes
Mobile Ministry Made Easy Handbook:
https://mobileministryforum.org/mobile-ministry-made-easy/

ConnectBox Specifications:

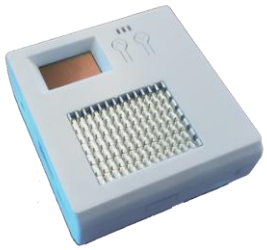
Storage:	Internal flash memory and/or external USB flash (FAT32/exFAT/NTFS/ext4)
LED's:	Green-Power on, Red-Battery charging, Orange – WiFi Operational
Switches:	Power On/Off-Forward & Backward push buttons
Power:	Internal 6400mAh Rechargeable 3.7 V Li-Ion Battery (95g) Model P only Removable K21700 or 18650 Li-Ion batteries of your choice and power Model R only 5VDC/2.0A external USB-C connection (EU or US) Model P & R 5VDC/3.0A external USB-C connection (EU or US) Model J 5VDC/2.0A external micro-USB connection (EU or US) Model S
Usable Battery Life:	6-8 hours on a full charge Model P & J (4 x 3200 MaH batteries) 10-14 hours with dual 5500maH K21700 full charge batteries Model R
Charge Time:	6 - 8 hours maximum from empty (unless deep discharge) Model P & R & J

Dimensions:	4.3 x 3.1 x 1.10 in (108 x 80 x 28 mm) Model P (Molded)
“	4.5 x 3.75 x 1.27 in (115 x 95 x 32 mm) Model R (3D Printed)
	2.9 x 3.1 x 1.10 in (73 x 80 x 28mm) Model S (3D Printed)
	5.1 x 4.3 x 1.13 in (130 x 110 x 31. mm) Model J (Molded)
Weight:	<7 ounce (196g) without external USB Model P
Processor	Allwinner H3 Quad-core 1.2Ghz Model P & S & R
	Raspberry Pi CM4 1.5Ghz Quad core BCM2711 Model J
Memory:	512MB DDR3 ARM
Wireless Standards:	IEEE 802.11n (b/g/n)
Frequency:	2.4-2.4835 GHz
	Support 64/128 bit WPA-PSK/WPA2-PSK (External Antenna Option Available)
Range:	Typically, 50m (160’) line-of-sight
Signal Rate:	Up to 300 Mbps
Operating:	Temp 15C - 40C, Humidity 0% ~75% non-condensing
Storage:	Temp -20C - 50C, Humidity 0%~75% non-condensing
Certification	CE



Battery Powered/Line Powered

Model P



Line Powered Only

Model S



Replacable K21700 or18650 Li-Ion Batteries

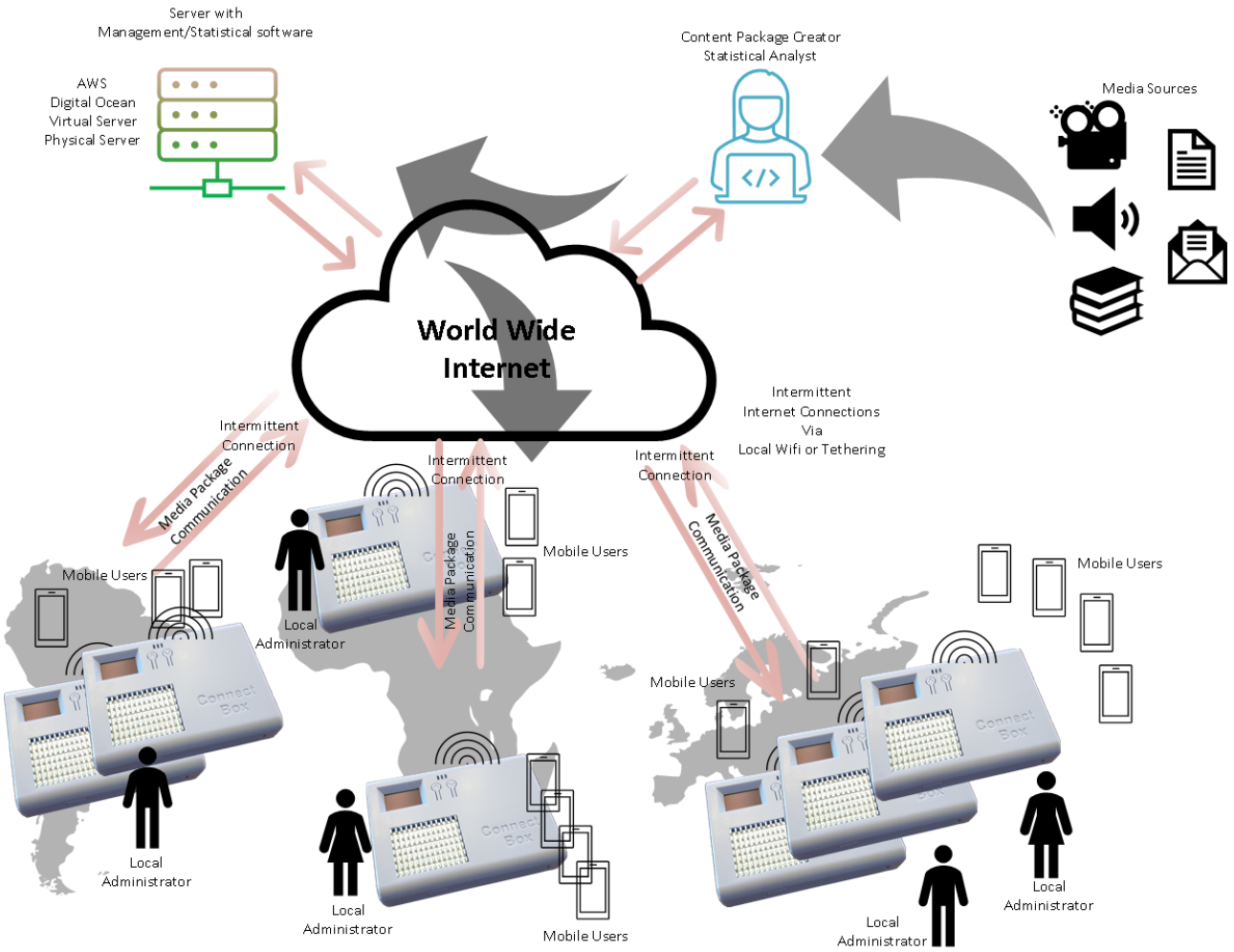
NEW Model R

NEW Model J



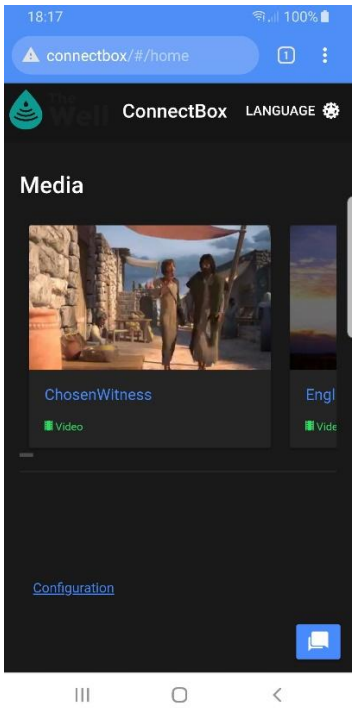
Replacable Li-Ion 4 x 18650 batteries
Raspberry Pi CM4 compute Module
Hdmi Out
Internal & External USB A

With a host Server management capability one can manage remotely the media contents and retrieve the statistical usage of the devices as well as control the device settings. This is a store-and-forward system that transacts when a ConnectBox is attached to the internet by tethering to a mobile phone or by Wi-Fi connection to the internet on an intermittent basis.

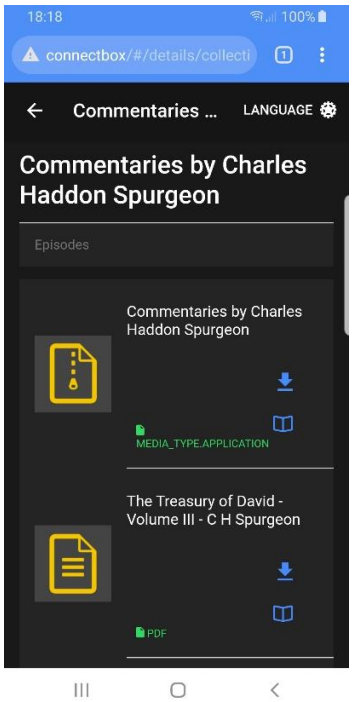


To utilize the central control platform we provide you with the instructions and code to create the platform on your own server or on a virtual server at AWS or Digital Ocean or other virtual provider. We provide an Ansible setup script that installs the server on Ubuntu 24.04, with secure (HTTPS) access set up automatically. You can then register your devices to this server address for management and control of the device in the field. A field manager will have to attach the ConnectBox to the internet from time to time either by Wi-Fi or by tethering to a mobile phone. The ConnectBox will then connect to the server, transact the media statistics and update any administrative controls; content packages prepared on the server can be downloaded from the box's administration page. You are then able to prepare reports of media viewing/downloading across all devices or by device. This then provides a management tool for evaluating the effectiveness of the media and the local administrator.

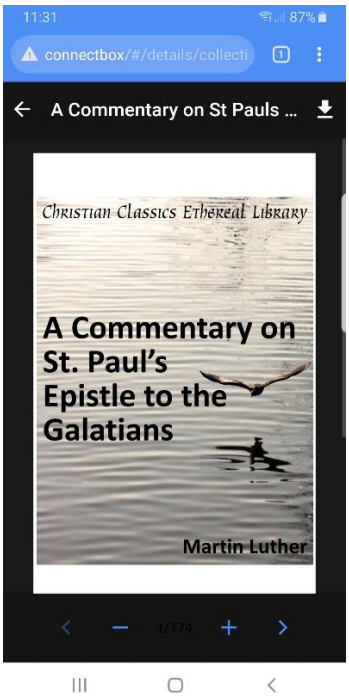
Views of the ConnectBox User Interface:



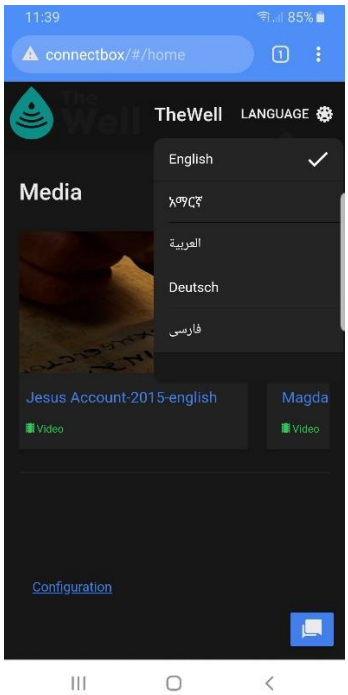
Video Rendering



Episodes

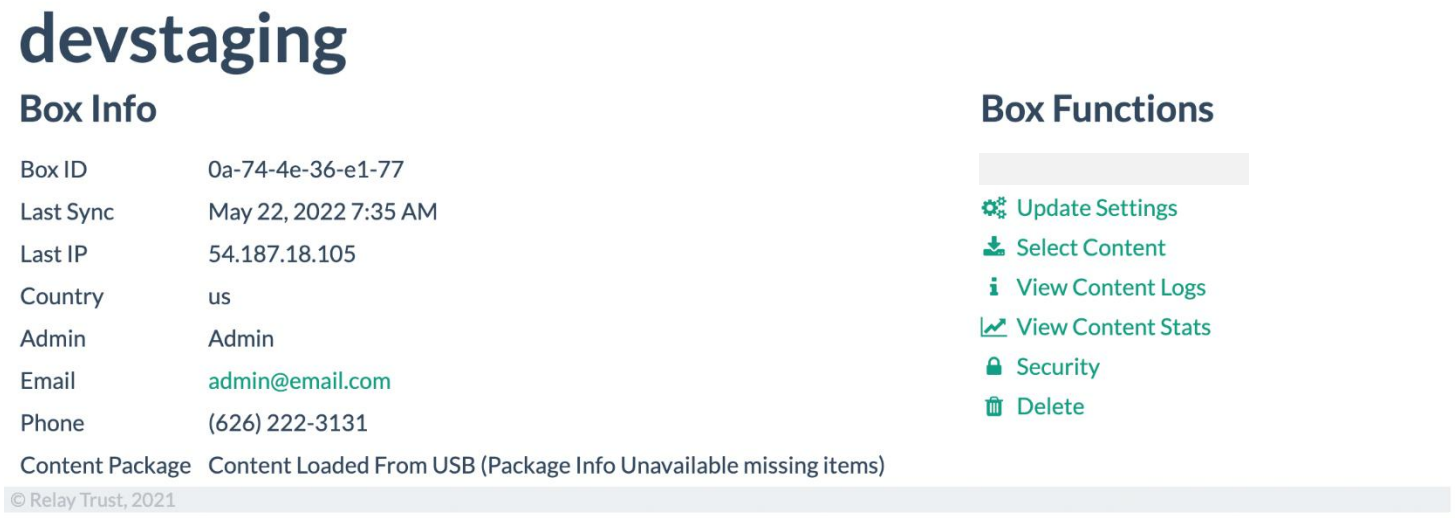


Built in PDF reader



Language Support

Views of the Control Platform Interface:



Status of a single ConnectBox.